



Amrita School of AI
COIMBATORE CAMPUS

UNIT 2 · SESSIONS 6 AND 7

Inner products and bases

Length, angle and components for complex vectors.

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25AID432 · Odd Semester 2026-27

SESSION 6

Length and angle for complex vectors

The inner product, the norm, normalisation, orthogonality.

The dot product of real vectors

$$\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2$$

Dotting a vector with itself gives its length squared:

$$(3, 4) \cdot (3, 4) = 9 + 16 = 25$$

Length 5, by Pythagoras.

A dot product of zero means the two vectors are perpendicular.

Try it on $\vec{v} = \begin{pmatrix} 1 \\ i \end{pmatrix} : 1 \cdot 1 + i \cdot i = 0$

A vector that is clearly not zero, with length zero. The real formula breaks for complex entries.

$$\langle \vec{u}, \vec{v} \rangle = \vec{u}^\dagger \vec{v} = \sum_k u_k^* v_k$$

The **inner product**. Conjugate the entries of the **first** vector, then multiply and add.

Now every length is real and positive

$$\langle \vec{v}, \vec{v} \rangle = \sum_k v_k^* v_k = \sum_k |v_k|^2$$

A sum of squared moduli: never negative, and zero only for the zero vector.

For $\vec{v} = \begin{pmatrix} 1 \\ i \end{pmatrix}$:

$$\langle \vec{v}, \vec{v} \rangle = 1 + (-i)(i) = 2$$

Which argument gets the conjugate?

- **This course, and all of physics**

Conjugate the **first**:

$$\langle \vec{u}, \vec{v} \rangle = \sum_k u_k^* v_k$$

- **Many maths textbooks**

Conjugate the **second**:

$$\sum_k u_k v_k^*$$

Same idea, opposite convention. Check which one a book uses before you copy a formula from it.

How the inner product behaves

Swap the order and you get the conjugate:

$$\langle \vec{u}, \vec{v} \rangle = \langle \vec{v}, \vec{u} \rangle^*$$

A number pulled out of the **second** slot comes out as it is; from the **first** slot it comes out conjugated:

$$\langle \vec{u}, c\vec{v} \rangle = c \langle \vec{u}, \vec{v} \rangle$$

$$\langle c\vec{u}, \vec{v} \rangle = c^* \langle \vec{u}, \vec{v} \rangle$$

$$\vec{u} = \begin{pmatrix} 1 + i \\ 2 \end{pmatrix}, \vec{v} = \begin{pmatrix} 3 \\ -i \end{pmatrix}$$

Conjugate $\vec{u}$, then pair up:

$$\begin{aligned} \langle \vec{u}, \vec{v} \rangle &= (1 - i)(3) + (2)(-i) \\ &= 3 - 3i - 2i \\ &= 3 - 5i \end{aligned}$$

The other order conjugates $\vec{v}$ instead:

$$\begin{aligned} \langle \vec{v}, \vec{u} \rangle &= (3)(1 + i) + (i)(2) \\ &= 3 + 5i \end{aligned}$$

The two are conjugates, as the rule says.

The length of a vector

$$\|\vec{v}\| = \sqrt{\langle \vec{v}, \vec{v} \rangle} = \sqrt{\sum_k |v_k|^2}$$

A **unit vector** has norm 1.

Worked: $\vec{v} = \begin{pmatrix} 3 \\ 4i \end{pmatrix}$.

$$\|\vec{v}\|^2 = |3|^2 + |4i|^2 = 9 + 16 = 25$$

So $\|\vec{v}\| = 5$.

Normalise: divide by the norm

$$\hat{v} = \frac{\vec{v}}{\|\vec{v}\|}$$
$$\begin{pmatrix} 3 \\ 4i \end{pmatrix} \rightarrow \begin{pmatrix} 3/5 \\ 4i/5 \end{pmatrix}$$

Now $\vec{w} = \begin{pmatrix} 1+i \\ 1 \end{pmatrix}$:

$$\|\vec{w}\|^2 = |1+i|^2 + 1 = 2 + 1 = 3$$

$$\hat{w} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1+i \\ 1 \end{pmatrix}$$

Every quantum state is a unit vector.

Normalising is the step that turns any non-zero column into a valid state. You will do it constantly in Unit 3.

Two ways to get the norm wrong

- **Squaring instead of taking modulus squared**

$$1^2 + i^2 = 0$$

Use $|v_k|^2 = v_k^* v_k$, which is never negative.

- **Dividing by the norm squared**

For $\begin{pmatrix} 3 \\ 4i \end{pmatrix}$, dividing by 25 gives a vector of length $1/5$.

Divide by $\|\vec{v}\| = 5$, then check the new norm is 1.

Inner product zero: orthogonal

$$\langle \vec{u}, \vec{v} \rangle = 0$$

For real vectors this is “perpendicular”. For complex vectors it is the definition.

Worked: $\vec{u} = \begin{pmatrix} 1 \\ i \end{pmatrix}, \vec{v} = \begin{pmatrix} 1 \\ -i \end{pmatrix}.$

$$\begin{aligned} \langle \vec{u}, \vec{v} \rangle &= (1)(1) + (-i)(-i) \\ &= 1 + i^2 = 0 \end{aligned}$$

Orthogonal.

Forget the conjugate and nonsense follows

- Without the conjugate

$$\begin{pmatrix} 1 \\ i \end{pmatrix} \text{ "with itself": } 1 + i^2 = 0$$

So $(1, i)$ would be orthogonal to itself.

$$(1, i) \text{ with } (i, 1): i + i = 2i$$

- With the conjugate

$$\langle (1, i), (1, i) \rangle = 1 + 1 = 2$$

Its length is $\sqrt{2}$, as it should be.

$$\langle (1, i), (i, 1) \rangle = i - i = 0$$

$$\vec{u} = \begin{pmatrix} 1 \\ 1 + i \end{pmatrix} \text{ and } \vec{w} = \begin{pmatrix} 1 - i \\ -1 \end{pmatrix}$$

Find:

- $\|\vec{u}\|$ and the unit vector $\hat{u}$
- $\langle \vec{u}, \vec{w} \rangle$
- Are $\vec{u}$ and $\vec{w}$ orthogonal?

Five minutes. Conjugate the first vector, and only the first.

$$\vec{u} = \begin{pmatrix} 1 \\ 1 + i \end{pmatrix} \text{ and } \vec{w} = \begin{pmatrix} 1 - i \\ -1 \end{pmatrix}$$

$$\|\vec{u}\|^2 = 1 + |1 + i|^2 = 1 + 2 = 3$$

$$\hat{u} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 + i \end{pmatrix}$$

$$\begin{aligned} \langle \vec{u}, \vec{w} \rangle &= (1)(1 - i) + (1 + i)(-1) \\ &= 1 - i - 1 - i = -2i \end{aligned}$$

Orthogonal.

`np.vdot` conjugates the first argument

```
import numpy as np

u = np.array([1, 1 + 1j])
w = np.array([1 - 1j, -1])
print(np.vdot(u, w))           # 0j           the inner product <u, w>
print(np.linalg.norm(u))       # 1.7320508... which is sqrt(3)
print(u / np.linalg.norm(u))   # the unit vector u-hat
print(np.dot(u, u))            # (1+2j)      no conjugate: not even real
```

`np.vdot` matches this course. `np.dot` does **not** conjugate, so never use it for a norm.

Before Session 7

With $\vec{a} = \begin{pmatrix} 2 \\ i \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} 1 \\ 2i \end{pmatrix}$:

find $\langle \vec{a}, \vec{b} \rangle$, $\langle \vec{b}, \vec{a} \rangle$, $\|\vec{a}\|$ and $\hat{a}$.

Are they orthogonal?

Check each answer with `np.vdot` and `np.linalg.norm`.

Next: bases, projection, and why the inner product gives you components for free.

SESSION 7

Bases, projection and components

One vector, many ways to describe it.

A set of vectors you can build everything from

A **basis** of $\mathbb{C}^n$ is a set of n vectors such that every vector is a linear combination of them, in **exactly one** way.

The standard basis of $\mathbb{C}^2$:

$$\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Unit length, and orthogonal in pairs

$$\langle \vec{e}_j, \vec{e}_k \rangle = \begin{cases} 1 & j = k \\ 0 & j \neq k \end{cases}$$

The bases quantum mechanics uses are all of this kind.

Three orthonormal bases of $\mathbb{C}^2$:

$$\begin{aligned} & \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ & \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \\ & \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \end{aligned}$$

Check that $\vec{b}_1, \vec{b}_2$ are orthonormal

$$\vec{b}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \vec{b}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

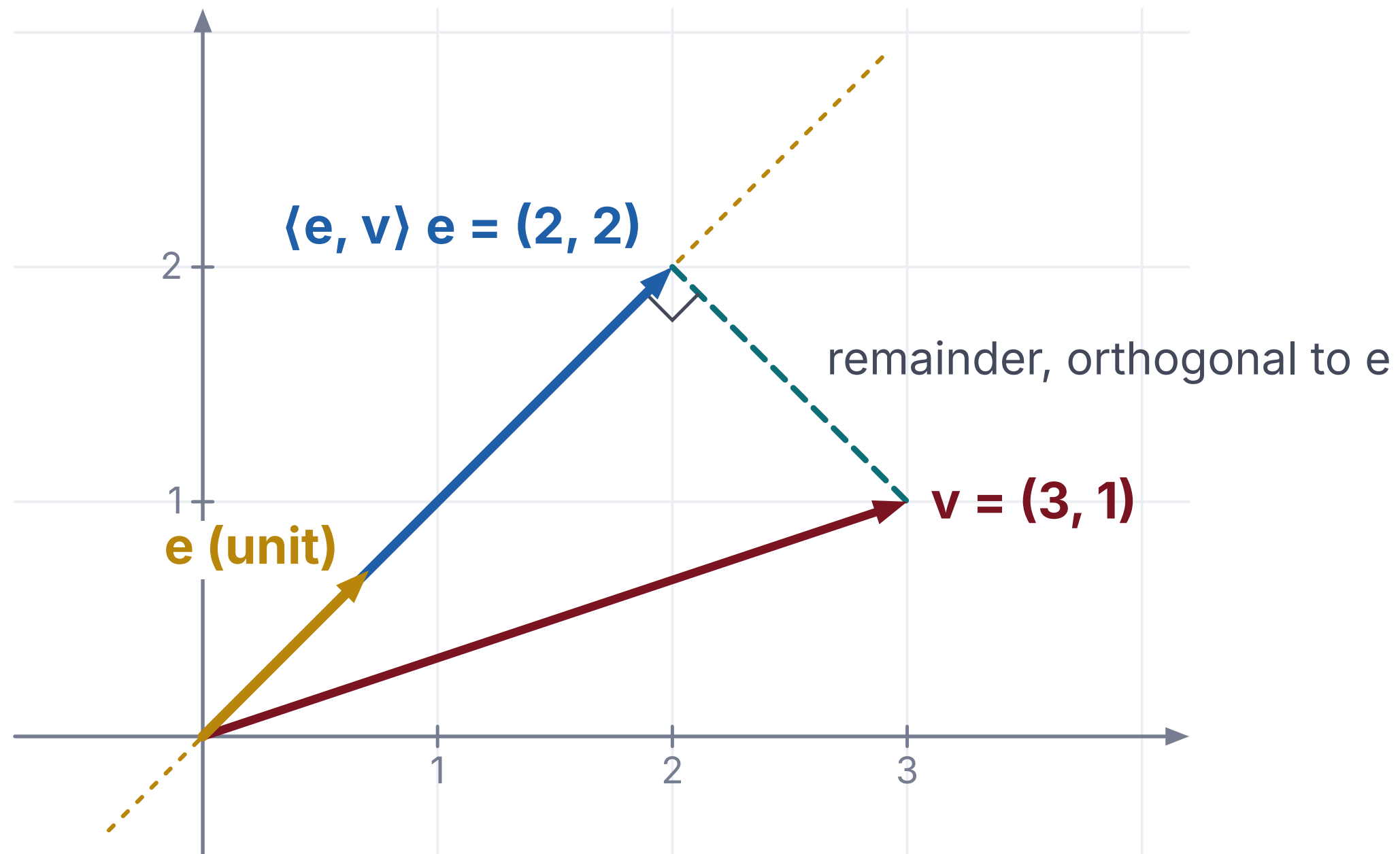
Three inner products to check. The $\frac{1}{\sqrt{2}}$ appears twice in each, giving $\frac{1}{2}$.

$$\langle \vec{b}_1, \vec{b}_1 \rangle = \frac{1}{2} (1 + 1) = 1$$

$$\langle \vec{b}_2, \vec{b}_2 \rangle = \frac{1}{2} (1 + 1) = 1$$

$$\langle \vec{b}_1, \vec{b}_2 \rangle = \frac{1}{2} (1 - 1) = 0$$

How much of $\vec{v}$ points along $\vec{e}$?



$$\text{Project } \vec{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \text{ onto } \vec{e} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

For a **unit** vector $\vec{e}$, the projection is

$$\langle \vec{e}, \vec{v} \rangle \vec{e}$$

and what is left over is orthogonal to $\vec{e}$.

$$\langle \vec{e}, \vec{v} \rangle = \frac{1}{\sqrt{2}} (3 + 1) = 2\sqrt{2}$$

$$2\sqrt{2} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

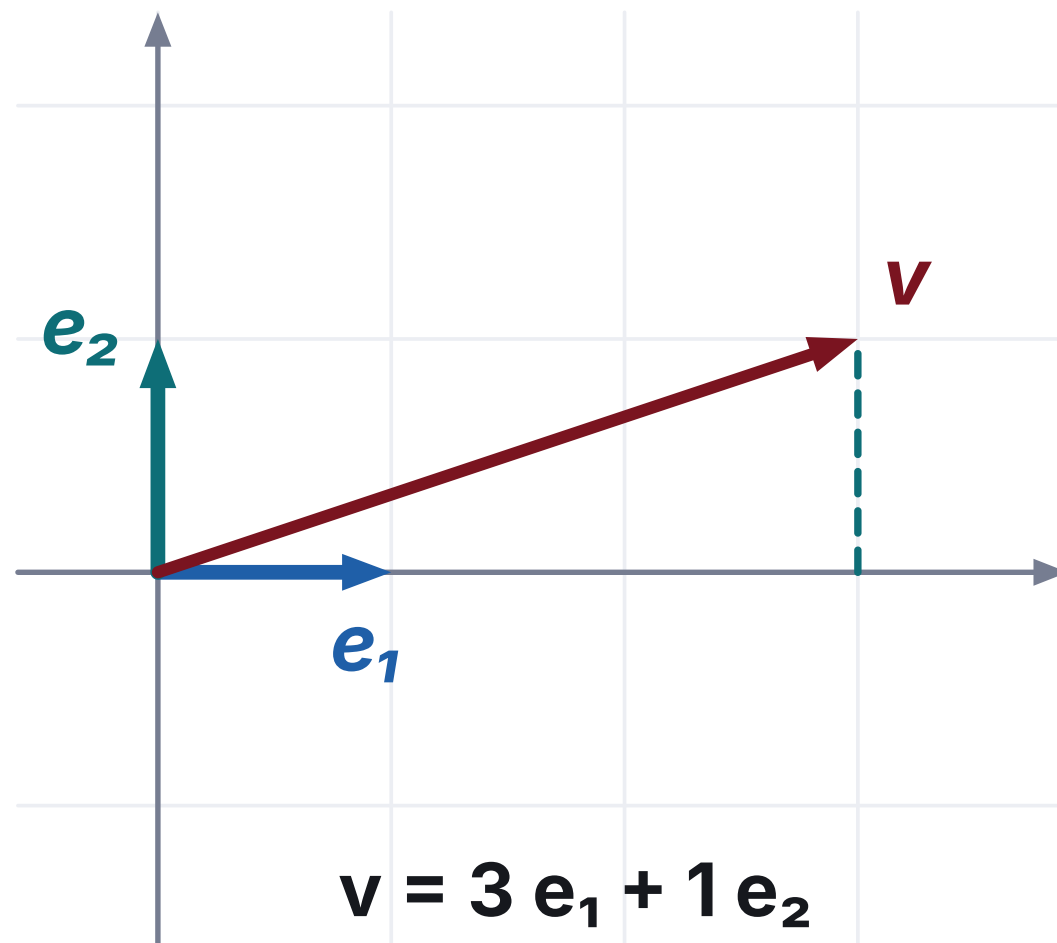
$$\text{Left over: } \vec{r} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \text{ and } \langle \vec{e}, \vec{r} \rangle = \frac{1}{\sqrt{2}} (1 - 1) = 0.$$

$$\vec{v} = \sum_k \langle \vec{e}_k, \vec{v} \rangle \vec{e}_k$$

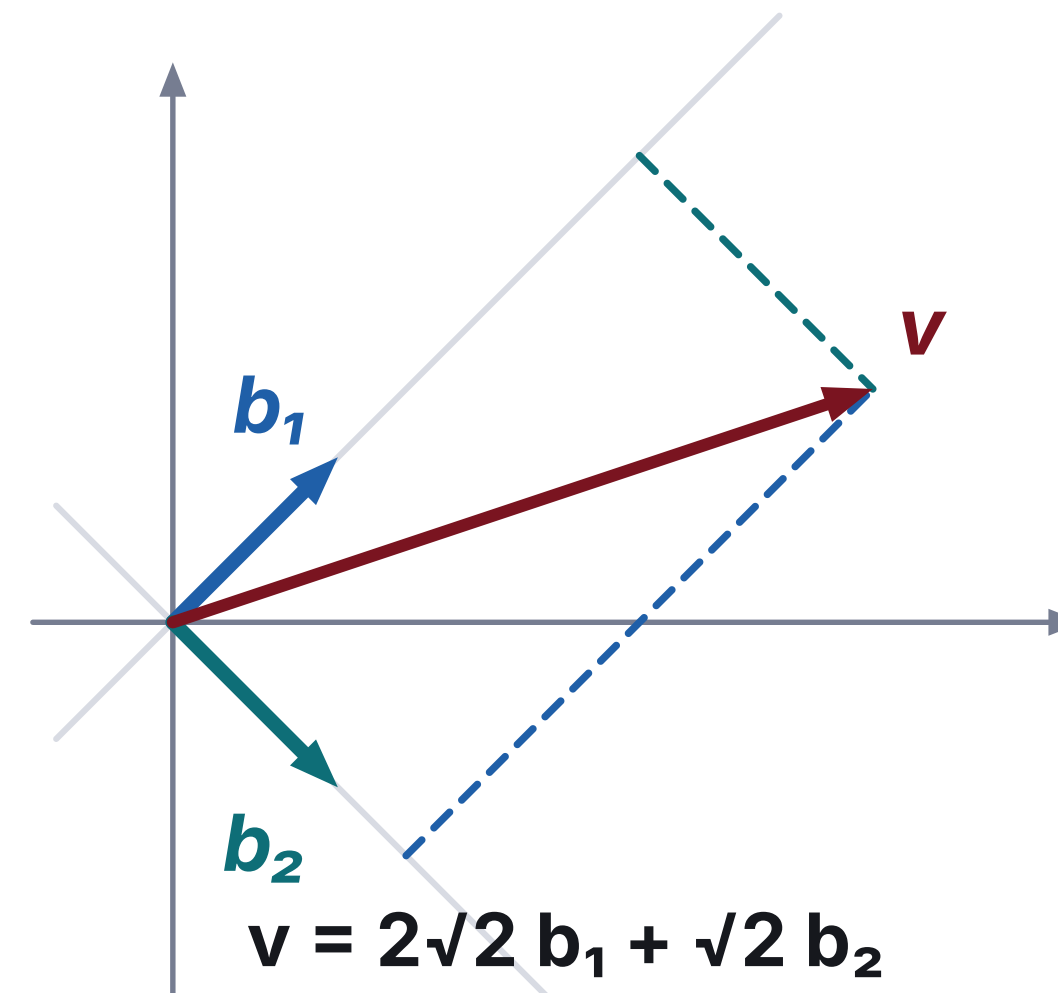
In an orthonormal basis, each component is one inner product. No simultaneous equations to solve.

Same arrow, different components

standard basis



basis b_1, b_2



Expand $\vec{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ in the basis $\vec{b}_1, \vec{b}_2$

$$c_1 = \langle \vec{b}_1, \vec{v} \rangle = \frac{1}{\sqrt{2}}(3 + 1) = 2\sqrt{2}$$
$$c_2 = \langle \vec{b}_2, \vec{v} \rangle = \frac{1}{\sqrt{2}}(3 - 1) = \sqrt{2}$$

Check by rebuilding $\vec{v}$:

$$2\sqrt{2} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \sqrt{2} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$
$$= \begin{pmatrix} 2 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \checkmark$$

Expand $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ in a complex basis

Basis: $\vec{f}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \vec{f}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}.$

The first slot conjugates, so $\vec{f}_1^\dagger = \frac{1}{\sqrt{2}} (1 \quad -i).$

$$c_1 = \langle \vec{f}_1, \vec{v} \rangle = \frac{1}{\sqrt{2}} (1 \cdot 1 + (-i) \cdot 0) = \frac{1}{\sqrt{2}}$$

$$c_2 = \langle \vec{f}_2, \vec{v} \rangle = \frac{1}{\sqrt{2}} (1 \cdot 1 + i \cdot 0) = \frac{1}{\sqrt{2}}$$

$$\frac{1}{\sqrt{2}} \vec{f}_1 + \frac{1}{\sqrt{2}} \vec{f}_2 = \frac{1}{2} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \checkmark$$

The shortcut has two conditions

- **Basis not orthonormal**

Take $(1, 0)$ and $(1, 1)$, and $\vec{v} = (3, 1)$.

The truth: $\vec{v} = 2(1, 0) + 1(1, 1)$.

Inner products give 3 and 4. Wrong.

- **Slots swapped**

The coefficient is $\langle \vec{e}_k, \vec{v} \rangle$, basis vector **first**.

Writing $\langle \vec{v}, \vec{e}_k \rangle$ gives the conjugate, which is wrong whenever it is complex.

The length does not depend on the basis

$$\|\vec{v}\|^2 = \sum_k |\langle \vec{e}_k, \vec{v} \rangle|^2$$

for any orthonormal basis $\{\vec{e}_k\}$.

For $\vec{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$:

standard: $3^2 + 1^2 = 10$

$\vec{b}_1, \vec{b}_2$: $(2\sqrt{2})^2 + (\sqrt{2})^2 = 8 + 2 = 10$

$$\sum_k |\langle \vec{e}_k, \vec{v} \rangle|^2 = 1 \text{ for a unit vector}$$

Each term is non-negative and together they sum to one: a probability distribution. In Unit 3 this becomes the Born rule.

$$\vec{v} = \begin{pmatrix} 3/5 \\ 4i/5 \end{pmatrix} \text{ in two bases}$$

Standard basis:

$$\left| \frac{3}{5} \right|^2 + \left| \frac{4i}{5} \right|^2 = \frac{9}{25} + \frac{16}{25} = 1$$

Basis $\vec{b}_1, \vec{b}_2$:

$$|\langle \vec{b}_1, \vec{v} \rangle|^2 = \frac{1}{2} \left| \frac{3}{5} + \frac{4i}{5} \right|^2 = \frac{1}{2}$$

$$|\langle \vec{b}_2, \vec{v} \rangle|^2 = \frac{1}{2} \left| \frac{3}{5} - \frac{4i}{5} \right|^2 = \frac{1}{2}$$

A different split, the same total.

Expand $\vec{v} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$ in the basis $\vec{b}_1, \vec{b}_2$

- Find $c_1 = \langle \vec{b}_1, \vec{v} \rangle$ and $c_2 = \langle \vec{b}_2, \vec{v} \rangle$.
- Check $|c_1|^2 + |c_2|^2 = \|\vec{v}\|^2$.
- Normalise $\vec{v}$. What are the two squared components now?

Five minutes. Recall $\vec{b}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\vec{b}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

$$\vec{v} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$$

$$c_1 = \frac{1}{\sqrt{2}}(4 + 2) = 3\sqrt{2}$$

$$c_2 = \frac{1}{\sqrt{2}}(4 - 2) = \sqrt{2}$$

$$18 + 2 = 20 = 4^2 + 2^2 \quad \checkmark$$

The norm is $\sqrt{20}$, so $\hat{v} = \frac{1}{\sqrt{20}} \begin{pmatrix} 4 \\ 2 \end{pmatrix}$ and every component shrinks by $\sqrt{20}$:

$$\frac{18}{20} = 0.9, \quad \frac{2}{20} = 0.1$$

All components at once

```
import numpy as np

b1 = np.array([1, 1]) / np.sqrt(2)
b2 = np.array([1, -1]) / np.sqrt(2)
v = np.array([4, 2])
c = np.array([np.vdot(b1, v), np.vdot(b2, v)])
print(c)                                # [4.24264069 1.41421356]    which is 3*sqrt(2), sqrt(2)
print(np.sum(np.abs(c) ** 2))           # 19.999999999999996      which is ||v||^2 = 20
print(c[0] * b1 + c[1] * b2)           # [4. 2.]               v rebuilt
```

Before Session 8

Expand $\vec{v} = \begin{pmatrix} 1 \\ i \end{pmatrix}$ in the basis $\vec{b}_1, \vec{b}_2$, and check $\sum_k |c_k|^2 = \|\vec{v}\|^2$.

Check that $\vec{f}_1, \vec{f}_2$ from today are orthonormal.

Qubit Workbook, Part B (linear algebra, 14 marks):
Sessions 4 to 9, due at the **end of week 10**.

Next: Hermitian and unitary matrices.